Reproducing LiTS Crossword Chains

A scoped reproduction with a crossing-consistency prompt intervention

FTEC5660 Individual Project 2 | Experimental run: 12-13 November 2026

Abstract

This project executes the released LiTS environment-grounded chain on 30 fixed 5x5 crosswords, with three stochastic trials per condition. The original prompt is compared with one appended instruction block that encourages cross-letter consistency. Both conditions use DeepSeek Flash with the same execution limits. Across 180 puzzle runs, final returned-state word accuracy changes from 26.22 +/- 3.08% to 18.56 +/- 1.71% (mean +/- sample SD across trial means), a difference of -7.67 percentage points. This is a model-substituted reproduction of a released multi-step agent behavior, not an exact replication of the paper benchmark.

1. Target and rationale

The source is LiTS: A Modular Framework for LLM Tree Search [1,2], one of the instructor-listed projects. The specific target is the Crosswords / Chain row in Table 3, not MCTS, BFS, or the whole framework. That row reports 6.67% puzzle accuracy and 10.33% partial word accuracy on 30 examples with Claude 3.5 Sonnet. These values provide context only: the original subset IDs and publication-time configuration were not established, and this project changes the model and fixes its own sample and budget.

Crosswords provide a small, inspectable environment with an exact evaluator. The selected program is agentic: each model call chooses an action, the environment changes the board, and the next call observes that changed state. It is not a single prompt evaluated once.

2. Reused architecture

Stage	Released component and role
Observe / decide	EnvGroundedPolicy formats the current board and clues; the model returns one placement such as h1. apple.
Validate / act	CrosswordsTransition validates the action and writes letters into the board, including crossing entries.
Loop / evaluate	EnvChain repeats up to 10 steps; the exact goal checker evaluates rows and columns. A successful goal check can stop early.

The chain, policy, crossword transition, data loader, and scoring code remain unchanged. The goal checker uses reference answers internally, including for early stopping. Those answers are not included in model prompts. This oracle stopping behavior is inherited from the original chain.

Controlled setup and intervention

3. Experimental design

Setting	Value shared by both conditions
Source revision	25e94d862cb4514c22d9a80a635e392461769df7
Sample	mini0505.json indices 0-29, selected before measured outputs; index 99 is a separate smoke check.
Replication	3 trials x 30 puzzles x 2 conditions = 180 runs. Condition order is reversed in trial 2. Up to 3 puzzle jobs run concurrently.
Model / endpoint	deepseek-flash (listed as DeepSeek-V4.1-Flash at setup); https://api.deepseek.com
Decoding	Thinking disabled; initial temperature 0.8; top_p 1.0; max output 128 tokens per call; one action per policy step.
Budget / retry	10 environment steps; original policy allows 3 attempts per action, with retry temperature increasing to 1.0. Actual API calls may exceed 10.
Primary endpoint	Returned-state word accuracy: correct rows plus correct columns, divided by 10. Puzzle success means all 10 are correct.

The dataset contains 156 puzzles. Its SHA-256 is 50590d5886fac5e0971f483df93668a8f3c638d8312a490655b869a93d569154. No local random seed can guarantee deterministic hosted API outputs. Trial IDs identify repeated runs rather than matched random-number streams.

4. The single experimental change

Only the modified condition appends this block to the original user prompt:

```
Before selecting the single action, check the crossing letters already on the board.
Prefer an unfilled entry with several known letters and a confident clue match.
Choose a word that matches both its clue and its existing crossing letters; avoid
overwriting a confident crossing answer with a conflicting guess. If one entry
is uncertain, choose another unfilled entry. Still output ONLY position. word.
```

Hypothesis: emphasizing existing crossing letters may reduce destructive placements and improve final word accuracy. The instruction does not enforce consistency in code, provide answers, add tools, increase the step limit, or add a separate verification call. Because early letters can be wrong, preserving them may also lock in mistakes.

A shared compatibility edit in lits/lm/openai_chat.py sends DeepSeek's documented thinking toggle [3], replacing the vLLM-style toggle only for this endpoint. Both arms use the same adapter. The original prompt has no system message in this configuration; the warnings about a missing system prompt do not indicate an absent user prompt.

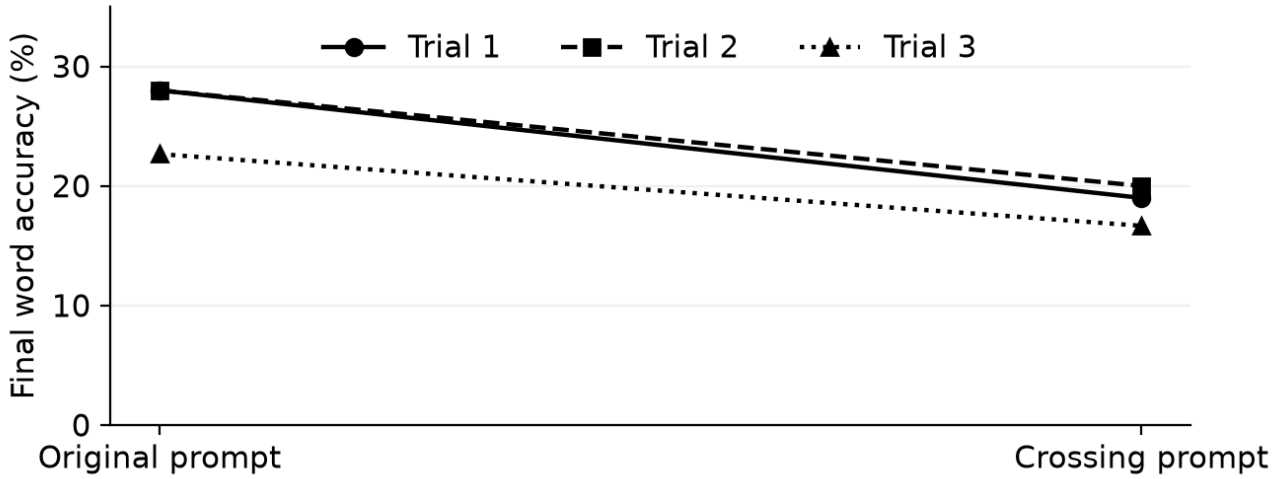
Measured results

5. Primary outcomes

Metric	Original prompt	Crossing prompt
Word accuracy (%)	26.22 +/- 3.08	18.56 +/- 1.71
Puzzle accuracy (%)	3.33 +/- 0.00	0.00 +/- 0.00
API calls / 30-puzzle trial	346.67 +/- 6.03	351.67 +/- 6.11
Input tokens / trial	102662.67 +/- 1831.61	131696.67 +/- 2191.78
Output tokens / trial	5928.33 +/- 704.71	7985.33 +/- 773.66
Seconds / puzzle	17.58 +/- 0.35	17.60 +/- 1.05
Jobs with terminal action error / trial	5.00 +/- 1.73	7.00 +/- 1.00

Entries are mean +/- sample SD across three trial-level values. Timing measures each puzzle's elapsed duration while jobs run concurrently, not serial benchmark latency. Token counts are API usage records; no monetary estimate is inferred from current pricing.

The process stopped after 173 completed jobs and resumed the seven unfinished jobs on 13 November. Incomplete traces are preserved separately; cost totals below cover completed jobs only. This interruption adds a time-of-run limitation.



Trial	Original word %	Modified word %	Difference (pp)
1	28.00	19.00	-9.00
2	28.00	20.00	-8.00
3	22.67	16.67	-6.00

The prompt intervention decreased mean final word accuracy by 7.67 percentage points in this sample. Across both arms there were 2,095 API calls, 703,078 input tokens and 41,741 output tokens. All 180 completed outcomes are included. No best-trial selection is used.

Sample variances of trial mean word accuracy (fraction scale) are 0.00094815 and 0.00029259. Three trials and a fixed convenience sample do not establish statistical significance or generalization.

Failure analysis and diagnostics

6. A state-accessor issue in the released code

When action validation exhausts its three attempts, the policy returns a terminal error step without next_state. EnvChain appends that step. EnvState.env_state reads only the last step and falls back to init_state if next_state is absent. Therefore the returned board can become blank even though earlier trajectory steps contain placements. The primary endpoint retains this released behavior; no fix was applied during the measured run.

The supplementary endpoint below scans the saved trajectory for the last valid board. It was added after this issue was observed and is explicitly post hoc. It diagnoses sensitivity to state handling; it is not a second intervention or a replacement primary metric.

Condition	Last-valid word %	Fallback jobs / 90	Invalid responses
Original	28.89 +/- 2.22	15	218
Modified	21.67 +/- 2.91	21	242

The last-valid-state difference is -7.22 percentage points, compared with -7.67 points for the primary returned-state endpoint. These endpoints answer different questions: output quality returned by the unmodified chain versus the quality of its last valid placement state. The diagnostic does not remove action failures from the primary analysis.

7. Inspectable examples

Largest observed improvement: trial 2, puzzle index 10

Original: 0.00%; modified: 70.00%. Calls: 10 versus 11. Raw evidence is in results/baseline/trial_1/puzzle_010/ and the matching crossing_prompt directory. These cases were selected after the experiment to illustrate variation, not to estimate the treatment effect.

Original returned board	Modified returned board
_____	BEBOP
_____	URENA
_____	FRIAR
_____	TONGA
_____	ORGAL

Largest observed regression: trial 1, puzzle index 13

Original: 100.00%; modified: 0.00%. Calls: 10 versus 15. Raw evidence is in results/baseline/trial_0/puzzle_013/ and the matching crossing_prompt directory. These cases were selected after the experiment to illustrate variation, not to estimate the treatment effect.

Original returned board	Modified returned board
BIPED	BFAET
URALI	UERLH
YAGER	YRREE
EDUCT	EMOCF
RESTY	INERT

Action validity is not semantic correctness: a five-letter, unfilled-position action may still contradict the clue or overwrite a crossing answer. A prompt can request better behavior without guaranteeing it.

Reproduction and evidence integrity

8. How to reproduce

Project repository: github.com/xwsyun/FTEC5660-LiTS-Reproduction

From the submitted repository root, use Python 3.12. Keep the API key in an ignored local .env file. The measured artifact can be audited without making API calls; running a new experiment requires API access.

```
python3.12 -m venv .venv
source .venv/bin/activate
pip install -r reproduction/requirements.txt
python reproduction/checks.py
unzip -n reproduction/results.zip -d reproduction
python reproduction/analyze.py --results reproduction/results
python reproduction/audit.py --results reproduction/results
# Optional new run; real API calls:
python reproduction/run.py --smoke --output reproduction/smoke_new
python reproduction/run.py --output reproduction/results_new
```

The direct dependency pins are in requirements.txt; the resolved environment is captured in requirements-lock.txt. The public repository packages raw traces in results.zip; extract it once before auditing. run.py records the fixed dataset hash, model settings, prompt intervention and source revision in config.json. Use a new output directory for new trials; existing completed jobs are skipped on resume.

Evidence	Purpose
PROTOCOL.md / DEBUG_DIARY.md	Pre-run scope and choices; setup difficulties and later diagnostic findings.
calls.jsonl / inferencelogger.log	Every successful API prompt/response, generation parameters, timing, and usage. No credentials.
trajectory.json / result.json	Environment steps and final returned-state score for every puzzle.
summary.json / per_puzzle.csv	Complete-set aggregation and per-puzzle outcomes.
audit.json	Independent board-score check, exact prompt reconstruction, response/trajectory alignment, and post-hoc last-valid-state diagnostic.

9. Verification and troubleshooting

Offline tests cover all 156 puzzles: an empty board scores 0, the reference board scores 1, and changing one letter reduces word accuracy to 0.8 because one row and one column become incorrect. A sentinel reference-answer string never enters the rendered prompt. The final audit reconstructs every actual prompt from board observations plus the condition-specific suffix and checks each saved score using an independent row/column comparison.

Setup required a focused hosted-API environment rather than all optional GPU/database packages. An unused asynchronous AWS dependency caused resolver backtracking and was removed. Eager imports still required IPython and other listed packages. A smoke-run harness bug assumed state.history existed; iterating EnvState fixed it before measured trials. The failed smoke artifacts are excluded from the measured dataset.

Discussion, reflection, and sources

10. What was and was not reproduced

The released chain can be executed end to end: actions depend on changing observations, trajectories are saved, and exact crossword outcomes can be recomputed. This reproduces a concrete multi-step behavior from the authors' code. It does not validate the original numerical result, the full paper, or the effectiveness of tree search. A different model, an unmatched subset, a later code revision and explicitly chosen decoding/step limits prevent a fair direct numerical comparison with Table 3.

The controlled within-project comparison estimates the effect of one appended prompt block under these settings. Its observed mean change is -7.67 percentage points in primary word accuracy. The modification also changes prompt length, so input-token overhead is part of the intervention cost. Retry frequency is an outcome, not a controlled equality of total API calls.

11. Reflection and limitations

The most useful reproducibility lesson is to inspect what a saved final state actually means. Successful execution, a fully populated board, and a correct puzzle are different events. The terminal-error fallback demonstrates how state access can affect an evaluation without any change in the model. Reporting both the unchanged endpoint and a labeled diagnostic makes that limitation visible.

The intervention is only a natural-language preference. It cannot enforce crossing consistency, and confidence is self-assessed. Stronger enforcement or repairing the state accessor would be worthwhile future experiments, but neither is mixed into this prompt-only comparison. The first 30 puzzles are a convenience sample; three repeats measure limited stochastic variability. Hosted model aliases and service behavior may change, and no API seed guarantees exact reruns.

AI assistance was used for implementation, debugging, analysis scripting and report drafting. The submission owner should review the evidence, adapt the reflection to their own understanding, and present the work personally. The raw evidence and executable checks are supplied so the claims can be inspected rather than accepted from prose alone.

References

[1] Xinzhe Li and Yaguang Tao. LiTS: A Modular Framework for LLM Tree Search. arXiv:2603.00631v1, 2026. Table 3. arxiv.org/html/2603.00631v1

[2] Authors' repository. github.com/xinzhel/lits-llm. Revision 25e94d862cb4514c22d9a80a635e392461769df7. Source files: lits/agents/chain/env_chain.py; lits/components/policy/env_grounded.py; lits/structures/env_grounded.py; demos/lits_benchmark/crosswords.py. Apache-2.0 license retained.

[3] DeepSeek API documentation, Thinking Mode. api-docs.deepseek.com/guides/thinking_mode/. Accessed 12 November 2026.